Combustion
Spring 2019

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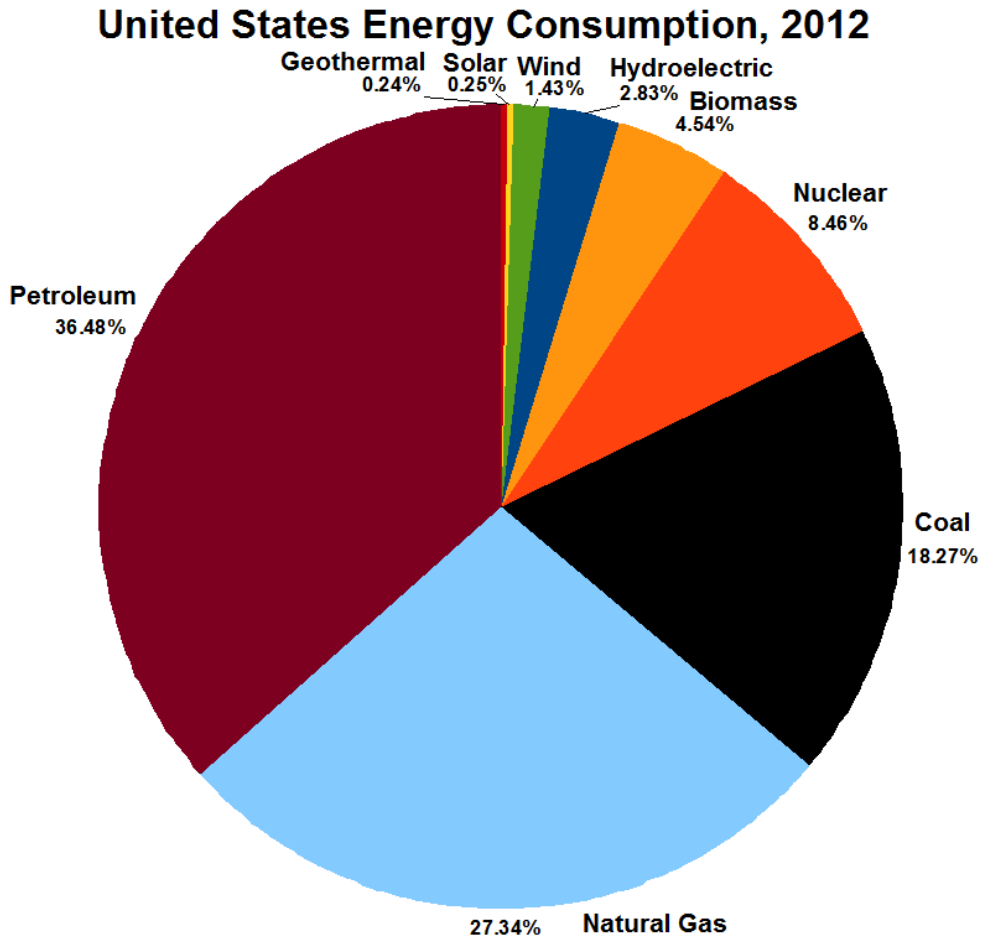
Lee Shau Kee Building B538

Introduction

- Prerequisites
  - Thermodynamics
  - Fluid Dynamics
  - Heat & mass transfer
  - Math

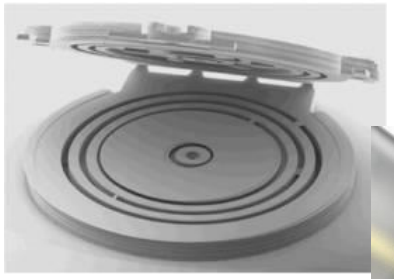
- Definition of combustion
  - Fuel oxidation with rapid heat release, (in the book)
  - Chemically reacting flows (more general)

Importance of Combustion Energy

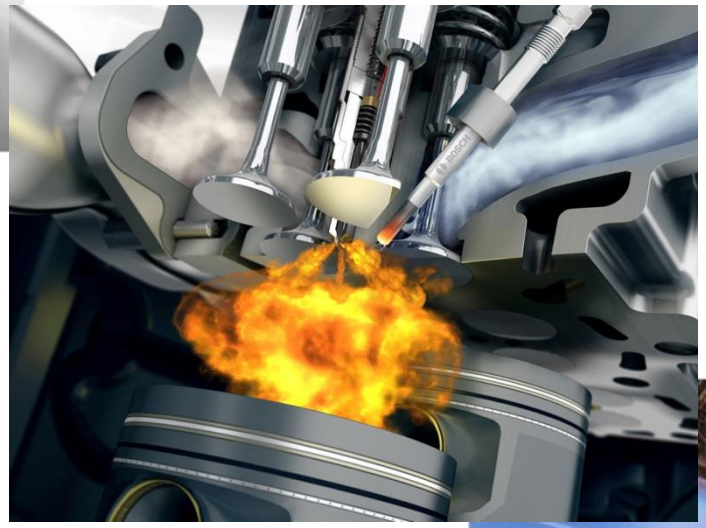


Importance of Combustion Energy

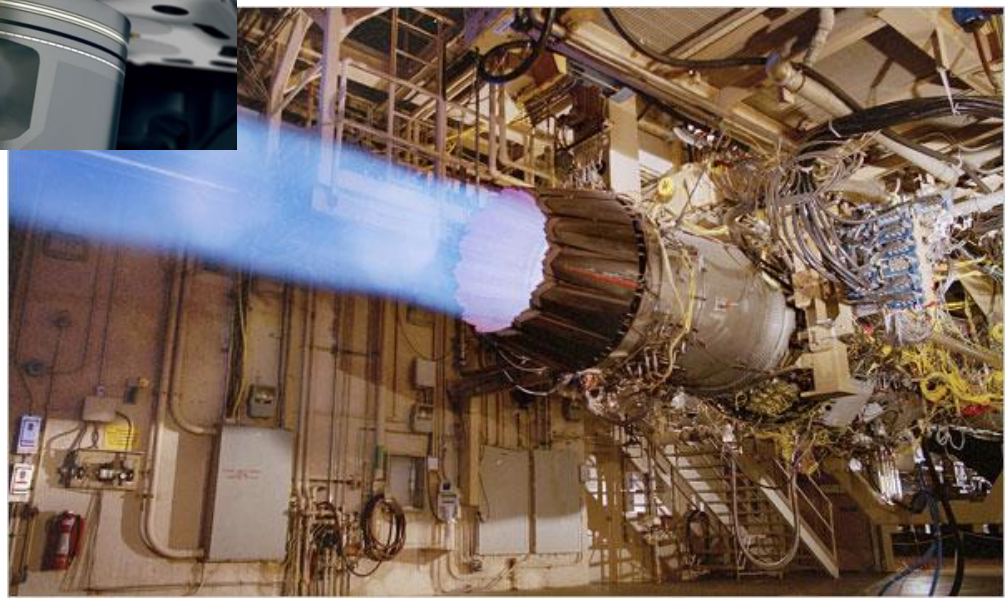
Micro-combustors (USC)



IC engines



Aerospace





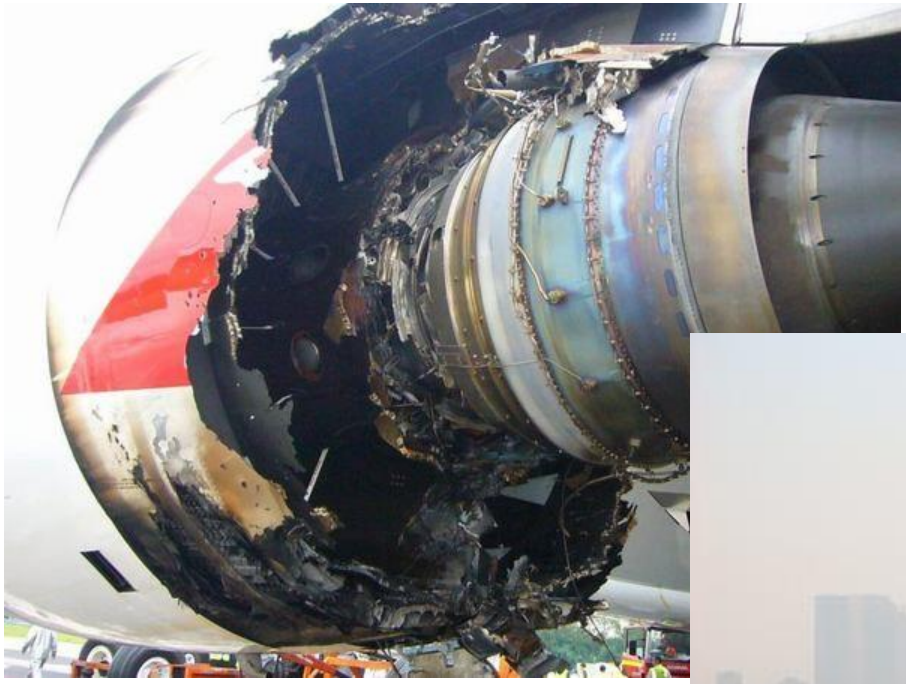






Adverse Effects

Fire safety



Emissions







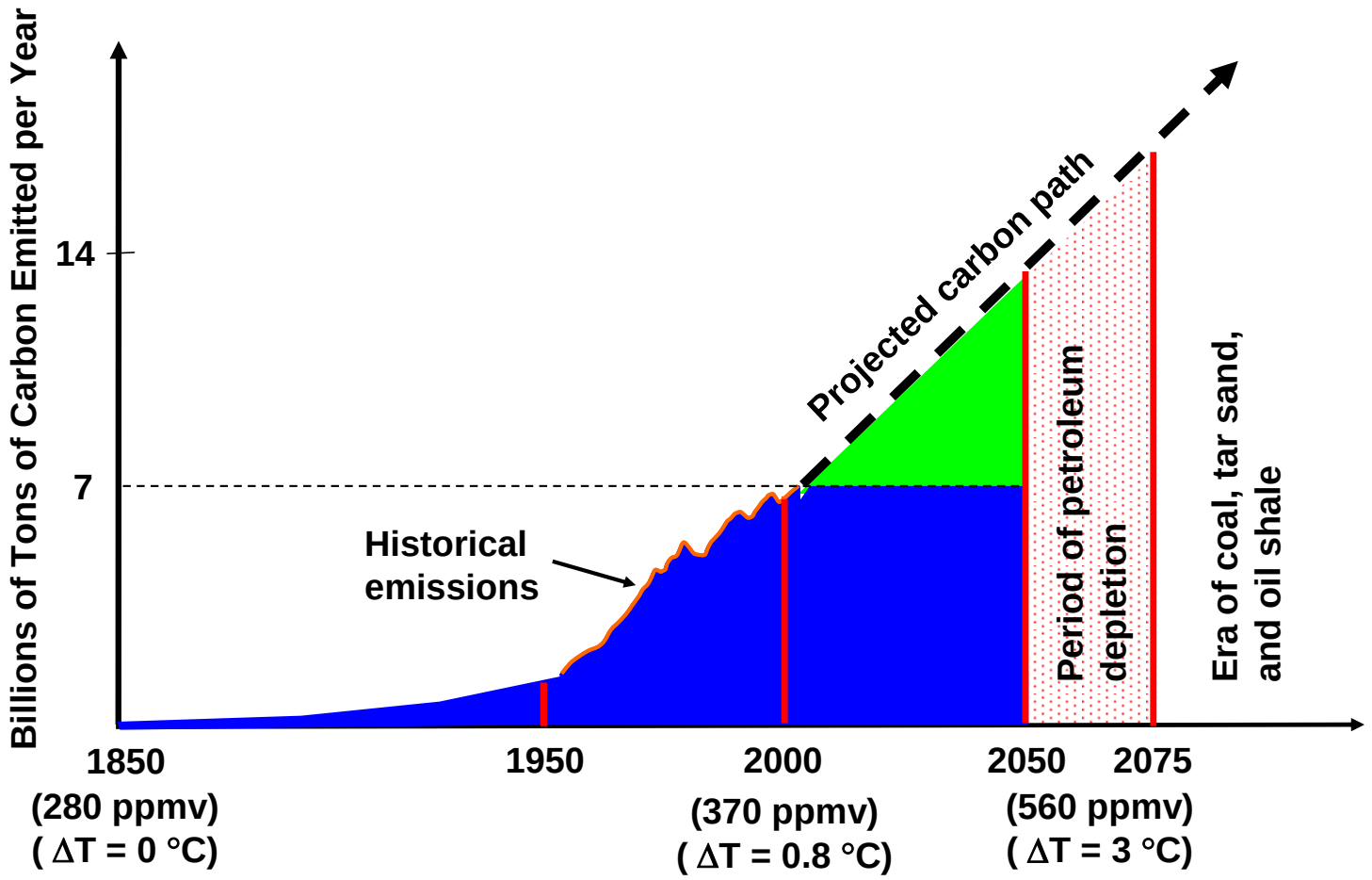


Soot



http://climateprogress.org/wp-content/uploads/2008/05/diesel.jpg

CO_2 Emission (Irreversible): The 50-75 Year Timeline



Climate Change

- Pollution continues to be a major concern; however, it is regional and mostly reversible on short time scales
- Global warming is reversible only on geological time scales, hence considerably more worrisome

Glacier National Park, Montana



1911

2000

Engine Efficiency

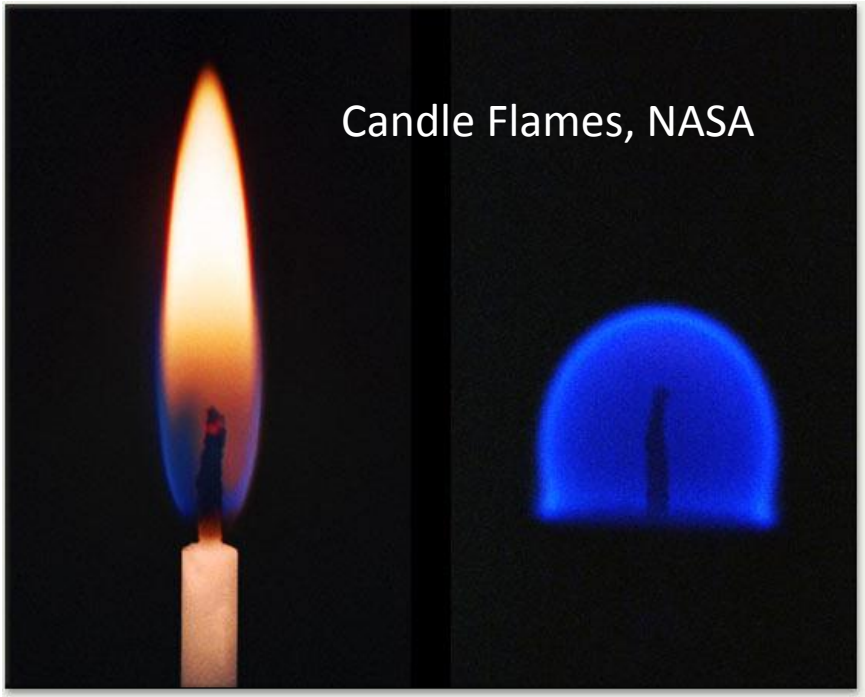
- Gasoline engines: < 30%



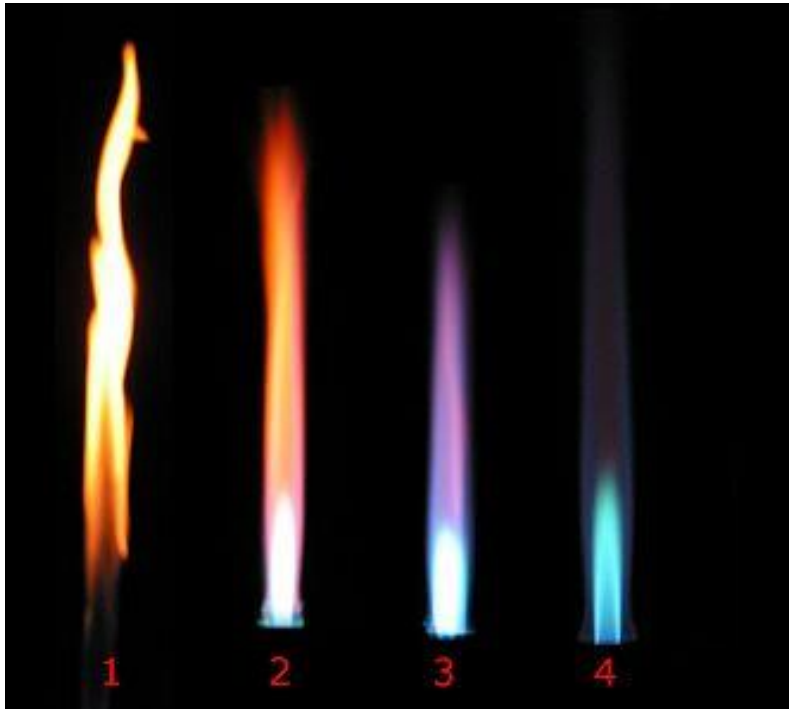
- Diesel engines: < 40%



Experimental Study



Bunsen Burner

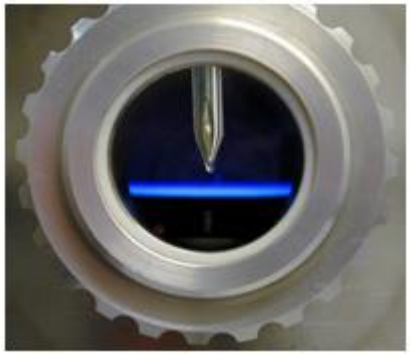




Shock tube, Stanford Univ



Counterflow flame



Flat flame burner

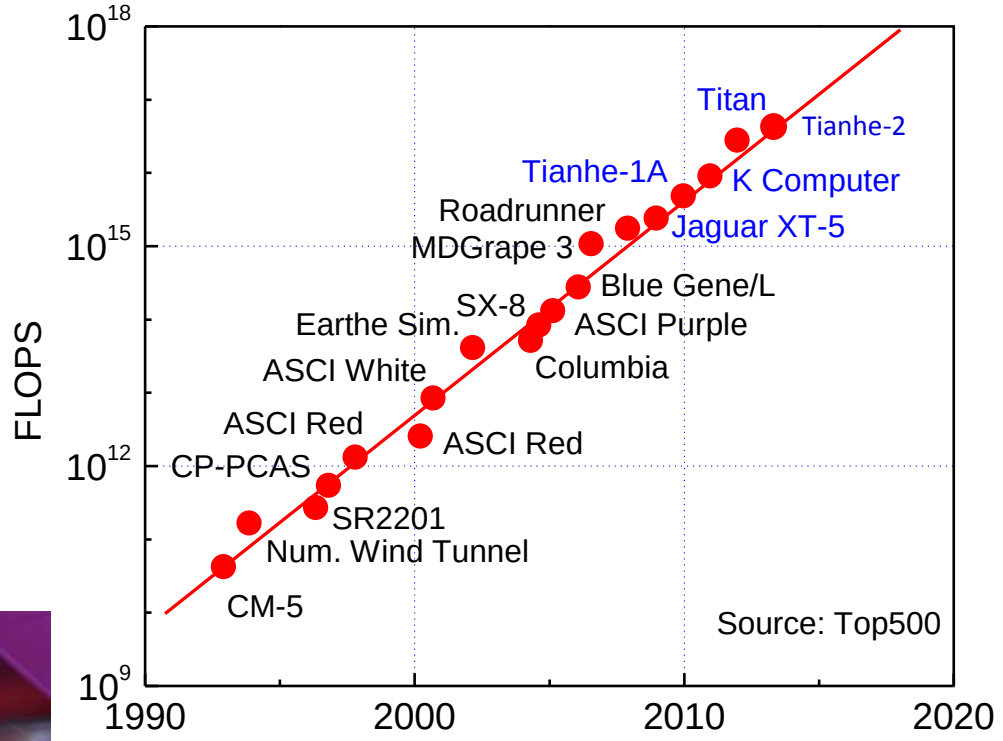


Flow reactor, Princeton

Computers and Numerical Combustion

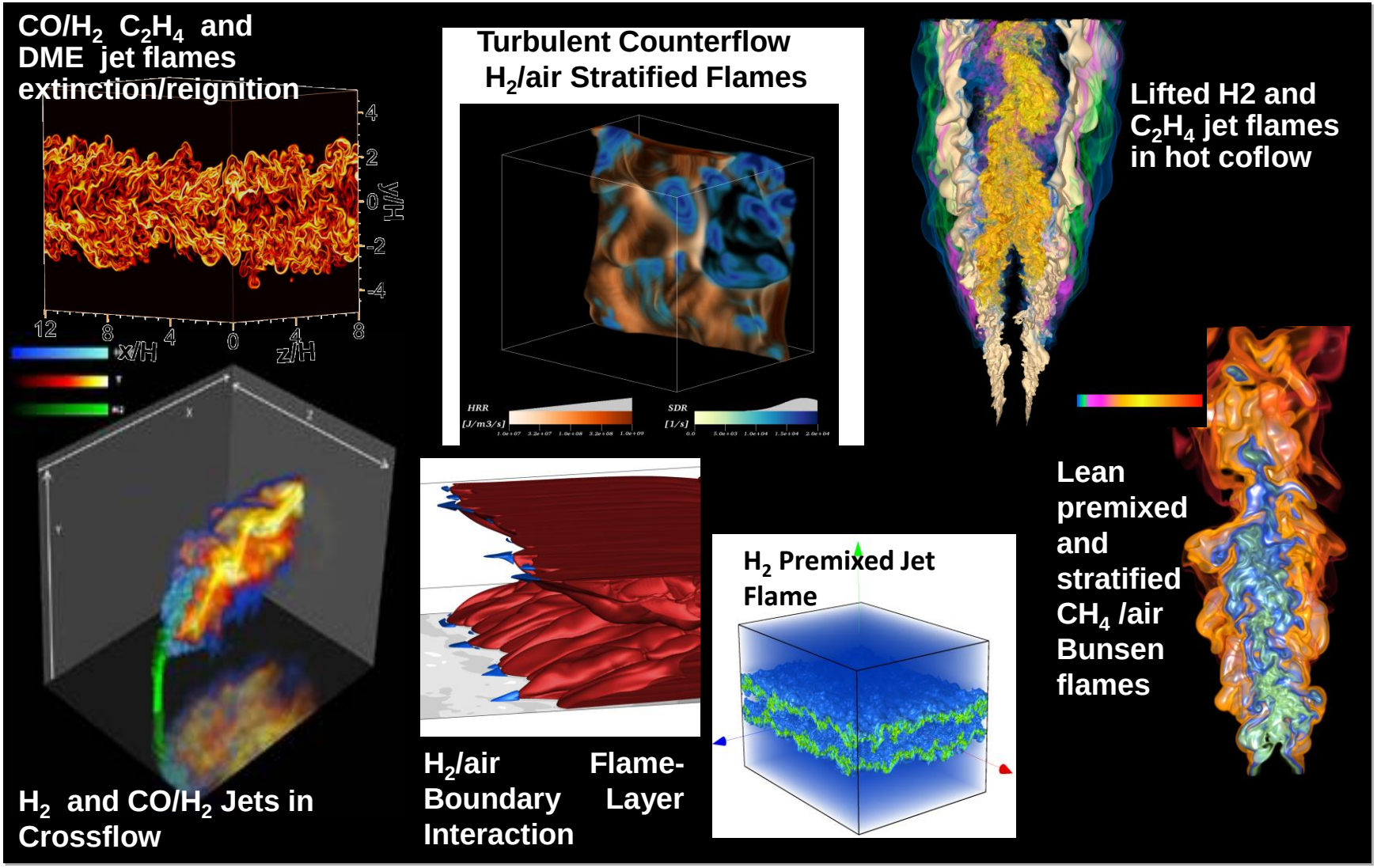




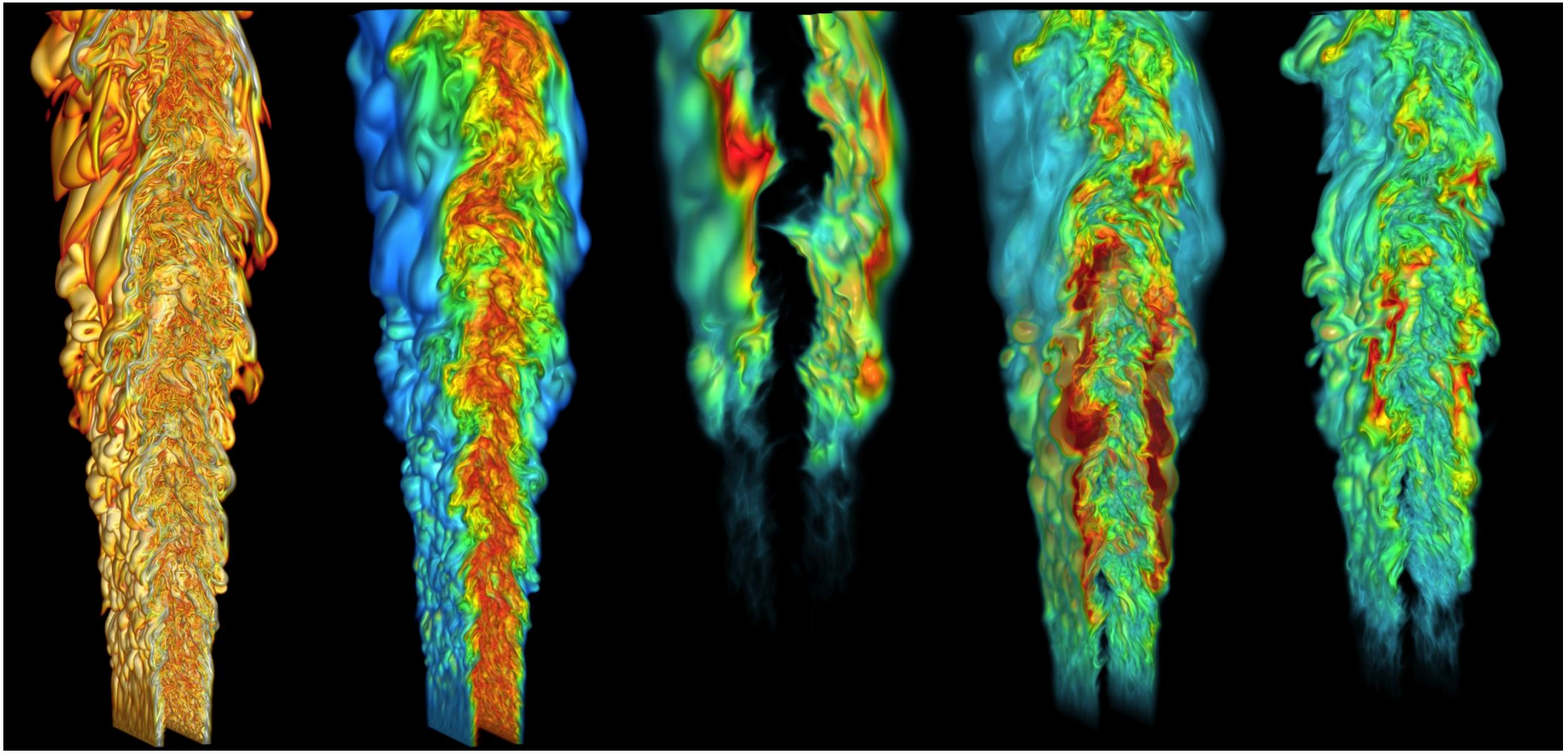


Tianhe-2, 3,120K cores, 55 PetaFLOPS

Recent DNS of Reacting Flows at Sandia

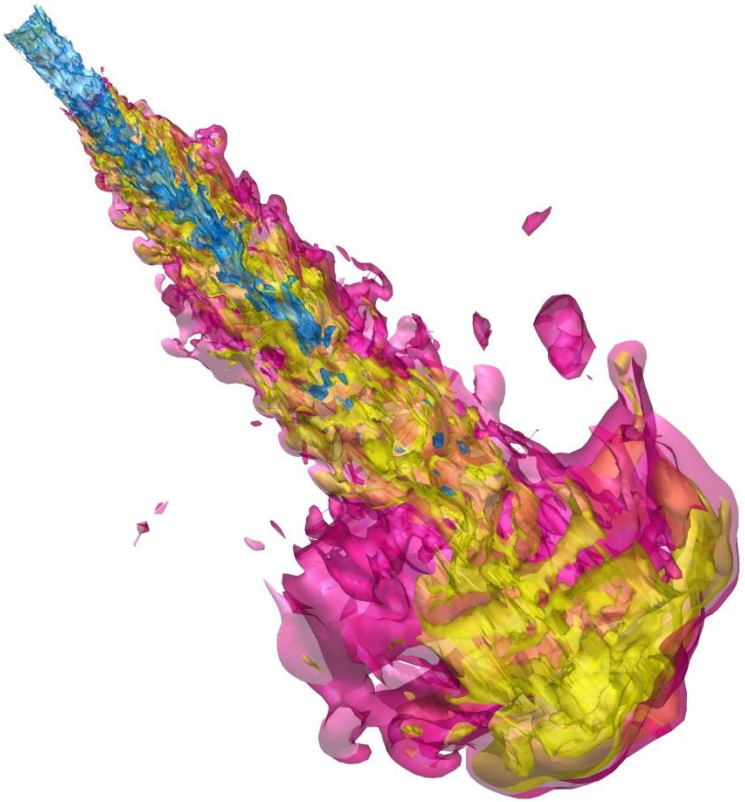


DNS of a Lifted Jet Flame

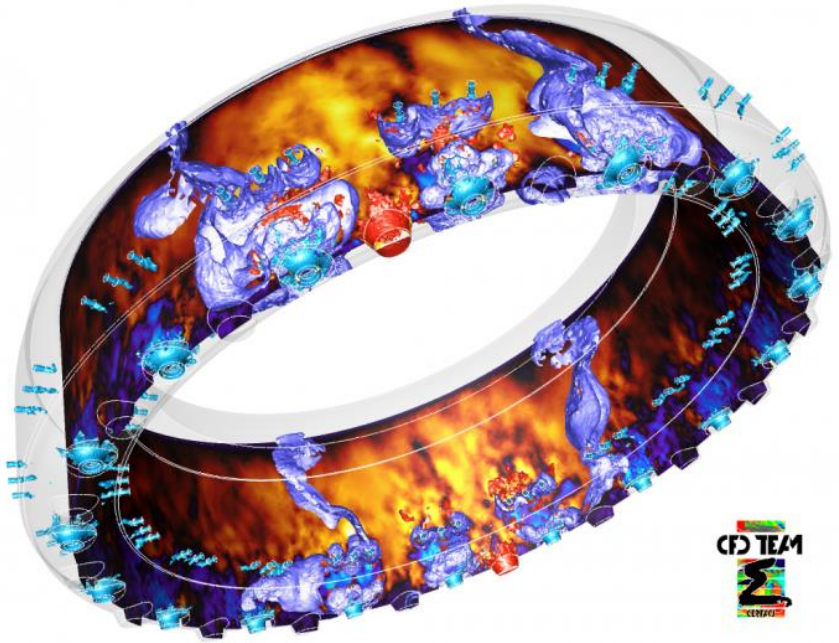


Combustion Research Facility, Sandia

Large Eddy Simulations



A diesel jet flame, Sandia



Gas Turbine Ignition, CERFACS, France

Emissions

- A major issue of the industrial world
- Types of pollutants
  - Oxides of nitrogen, NOx (NO, NO_2)
  - Sulfur oxides, SOx (SO_2, SO_3)
  - Unburned hydrocarbons, UHC
  - Particulates, soot
  - Heavy metals (Hg, Pb etc)
  - CO_2, green house gases
  - Carbon Monoxide, CO
  - Other

- Air pollution legislation in US, Clean Air Act (CAA), EPA
  - Official website: http://www.epa.gov/air/caa/
  - History of CAA: http://www.epa.gov/air/caa/caa_history.html

Topics to Study in Combustion

- Combustion
  - A general definition: chemically reacting flows
    - Thermodynamics, with heat release
    - Chemical kinetics
    - Fluid mechanics, with heat and mass transfer
    - Need to know conservation of:
      mass & elements, momentum, energy
- Approaches to study
  - Experiments
  - Analytical analysis
  - Numerical simulations

Detailed List of the Topics

- Thermo-chemistry
- Chemical kinetics
- 0-Dimensional chemical reactors
- Heat and mass transfer
- Laminar flames
  - Premixed flames
  - Non-premixed flames
- Droplet combustion
- Detonation
- Introduction to turbulent combustion
- Pollution control

Grading Policy (tentative)

- Homework & projects (30%)
- Quizzes (no make-up) & midterms (40%)
- Final exam (30%)

- Each must turn in one's own homework and project.
  - Discussions on general topics are strongly encouraged
  - Detailed problems should be solved independently
  - Identical or nearly-identical submissions are not allowed

Numerical Projects

- Based on the CHEMKIN package
  - www.reactiondesign.com
- Programming languages
  - MATLAB (recommended)
  - Any other languages, e.g. C/C++, Fortran
- MATLAB is strongly suggested for numerical projects